

Skills Summary

Sorting Shapes by What Makes Them That Shape

Use this reference while completing your worksheet or when studying.

Skill 1 — Counting Sides and Corners

A **side** is a straight edge. A **corner** is the point where two sides meet.

In every flat shape with straight sides, the number of sides equals the number of corners. Triangle: three sides, three corners. Square and rectangle: four and four. Hexagon: six and six. Circle: no straight sides and no corners — one curved edge all the way round.

How to count without losing your place: start at one corner, touch each side once, and stop when you get back to where you started.

Example: Which has more straight sides, a rectangle or a triangle?

Step 1. The question asks about sides, so count the straight edges of each shape and then compare the two numbers.

Step 2. Rectangle: top, bottom, left, right — that is four. Triangle: three.

⇒ $4 > 3$, so the rectangle has more.

Try it: Which has more straight sides, a hexagon or a triangle?

check: 6, the hexagon, $6 > 3$

Skill 2 — Colour and Size Do Not Matter

Defining attributes are the things you can count or check: number of straight sides, number of corners, whether the shape is closed, whether the edges are straight or curved. These decide what the shape IS.

Non-defining attributes are colour, size, and which way the shape is turned. A shape keeps its name when you make it bigger, paint it another colour, or stand it on a point.

Test to use: if you are not sure, count the sides. If the counts match, the shapes have the same name.

Example: A tiny triangle and a huge triangle — are they both triangles?

Step 1. Size is not something you count on the shape itself, so ignore it and count the sides on each shape instead.

Step 2. Tiny one: three sides. Huge one: three sides.

⇒ $3 = 3$, so yes — both are triangles.

Try it: A red square and a blue square — count the corners of each. Are they both squares?

check: 4 = 4, yes

Skill 3 — Sorting Shapes into Groups

Sorting means picking one rule and using only that rule. Good rules to sort by: “has four straight sides”, “has three corners”, “has no straight sides”.

Bad rules: “is red”, “is big”, “looks like a box”. Those change from shape to shape without changing what the shape is.

After sorting, count each group — then you can say which group has more.

Example: Sort a square, a rectangle, a trapezoid, a triangle and a circle by “has four straight sides”. Which group is bigger?

Step 1. Pick the one rule and test each shape against it, then count how many landed in each group.

Step 2. Four straight sides: square, rectangle, trapezoid — that is three. Not four sides: triangle, circle — that is two.

⇒ $3 > 2$, so the four-sided group is bigger.

Try it: Out of six shapes, two have three corners and the rest do not. Which group is bigger?

check: $2 < 4$, the “not three corners” group

Skill 4 — Putting Shapes in Order

To put shapes in order, turn each shape into a number first — how many sides, or how many corners — and then order the numbers.

Fewest to most means smallest number first. **Most to fewest** means largest number first.

Remember the circle: it has zero straight sides, so it comes first in a fewest-to-most list and last in a most-to-fewest list.

Example: Put a square, a triangle and a hexagon in order from the fewest corners to the most.

Step 1. Order words like “fewest” are about numbers, so count each shape first and then sort the numbers, not the pictures.

Step 2. Square: four. Triangle: three. Hexagon: six. Smallest first: three, then four, then six.

⇒ **3, 4, 6**

Try it: Put a pentagon, a triangle and a square in order from the fewest corners to the most.

check: **3, 4, 5**